

Chapter One

DC AND AC DEFLECTION METER MOVEMENTS

For many decades the basic de meter movement has been used as the readout device in electronic instruments, even in many instruments that measure AC values. Digital readout devices are common today, but until the mid-1980s they were too expensive for most common instruments. Even today; when digital displays are commonplace and low cost, analog meter movements are still used extensively. For some applications the analog meter is even preferred over the digital (e.g., in finding a peak or nul-condition, or where a zero-center application is needed); The two most common de meter movement configurations are the **d' Arsonval** and the **taut-band** designs. Both are examples of the **permanent magnet moving coil (PMMC)** galvanometer, and both work on the same fundamental principle as the de motor.

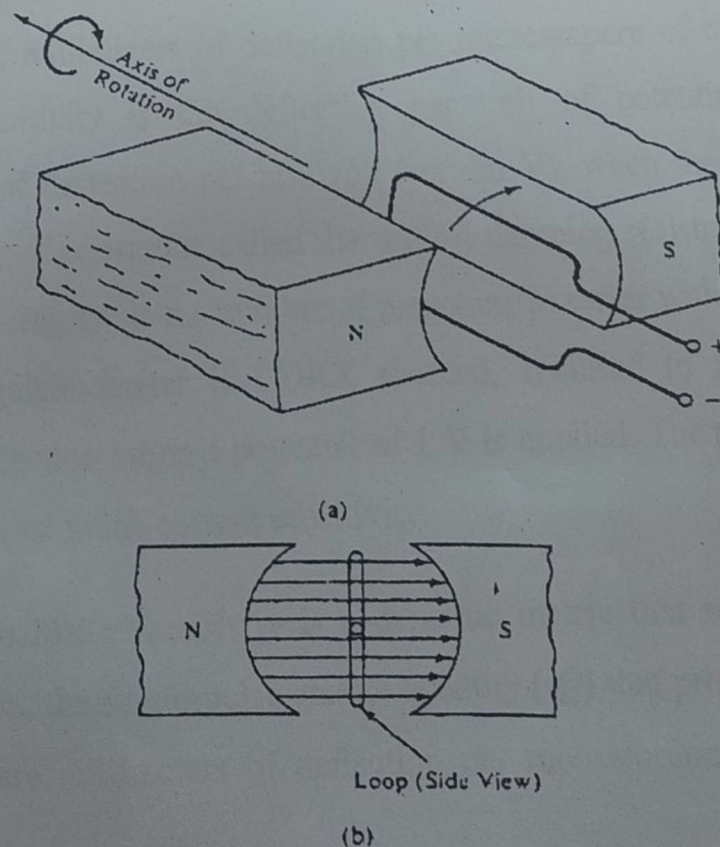


Fig. (1-1) (a) Construction of a d' Arsonval meter movement. This meter uses a loop of wire rotating in magnetic field, much like an electrical motor. (b) Side view of the d' Arsonval meter movement.

A simplified view of the d' Arsonval meter movement is shown in Figs. 1-1(a) and 1-1(b). A moving coil of wire is mounted in the magnetic field between the poles of a permanent magnet. A current flowing in a wire generates a magnetic field. The polarity of the field is determined by the direction of the current flow, while the strength of the field is determined by the magnitude of the current.

The coil in a PMMC galvanometer is mounted so that it can rotate in the space between the magnetic poles. A current in the coil creates a magnetic field that either aids or opposes the field of the nearby magnetic poles. A current flow in one direction causes a clockwise rotation, whereas a current flow in the opposite direction causes a counterclockwise rotation. The amount of rotational position change is proportional to the magnitude of the current.

The sensitivity of a galvanometer can be defined in several different ways: current sensitivity (S_I), voltage sensitivity (S_V), megohm sensitivity (S_R), and ballistic sensitivity (S_Q). The current sensitivity is the ratio of the deflection of the meter movement to the current that produced that deflection, for example, millimeters of deflection per microampere of current ($\text{mm}/\mu\text{A}$). Voltage sensitivity is the deflection per unit of potential, for example, millimeters of deflection per millivolt (mm/mV), when the meter is shunted with a value of resistance called the critical damping resistance (CDRX). The megohm sensitivity is the number of megohms in series with the galvanometer, when the galvanometer is CDRX shunted, required to produce one scale division deflection when a potential of 1 V is applied. The megohm sensitivity is usually equal to the current sensitivity.

The ballistic sensitivity is a dynamic metric that relates the maximum deflection to the single-pulse charge quantity (Q) that produced the deflection. The units are millimeters of deflection per microcoulomb of electric charge ($\text{mm}/\mu\text{C}$).

The D'Arsonval Meter Movement

The basic d' Arsonval meter movement is shown in Fig . . 1-2. A side view of the meter, without the permanent magnet, is shown in Fig.1-2(a), while a frontal view with the magnet in place is shown in Fig. 1 -2(b). The coil in Fig. 1-2(a) is wound on an armature or bobbin, which is mounted on a pair of jewel bearings to reduce friction. The assembly is shown (simplified) frontally in Fig.1,-2(b).

When a current flows in the coil, the armature assembly deflects clockwise [as shown in Fig. 1-2(b)] an amount that is proportional to the strength of the current. The amount of deflection can be marked off in units of current on the dial scale. The coiled pivot spring serves to dampen the pointer movement and to return the pointer to the zero position when the current flow in the coil ceases. The travel of the pointer is limited by high- and low-end mechanical stops just beyond the zero and full-scale limits printed on the dial scale. The wires to the coil are flexible and are given just enough slack that they will not be stretched anywhere in the pointer's normal range of travel.

The Taut-Band Meter Movement

The taut-band meter movement is essentially the same as the d' Arsonval movement, except for the manner in which the armature coil is mounted (see Fig.1-3). In the taut band movement the armature is suspended from fixed supports on a stretched (i.e., "taut") rubber band. The band is twisted as the armature rotates, so no restoring force from a pivot spring is needed. The taut-band meter movement has two principal advantages over the older d'Arsonval movement: greater sensitivity and more durability.

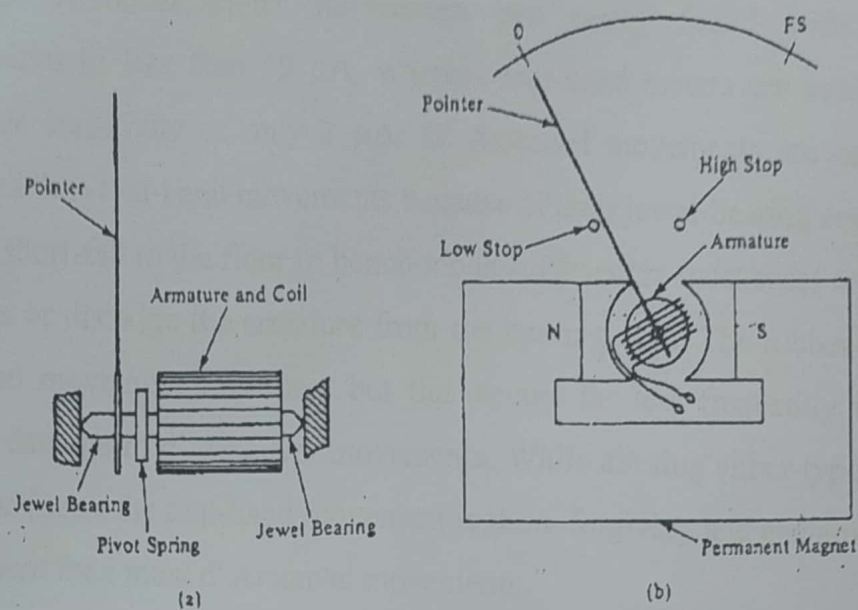


Fig. (1-2) (a) Construction of the coil and pointer assembly in an Arsonval meter movement, (b) usual configuration for a d'Arsonval meter movement.

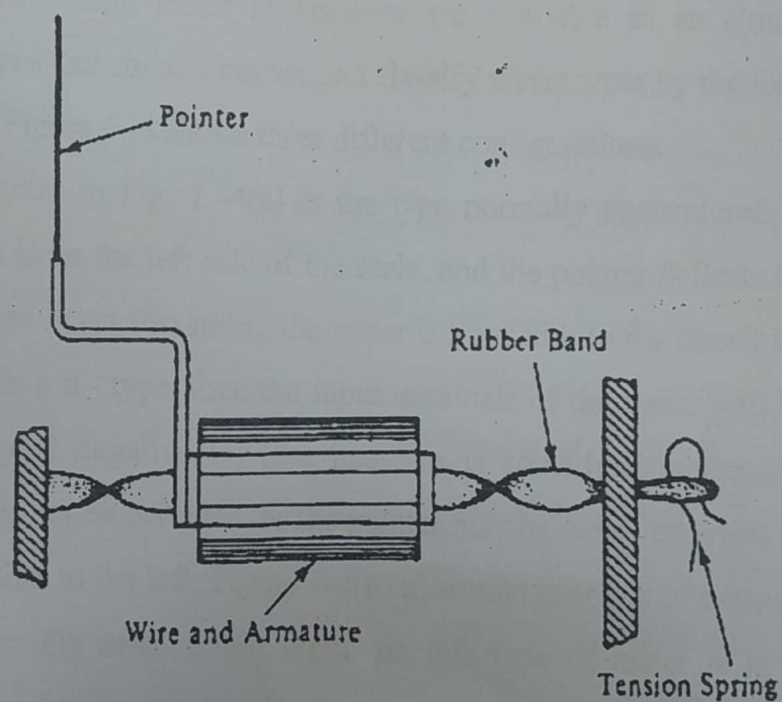


Fig. (3) Construction of taut-band meter movement.

D' Arsonval meter movements are rarely found with full-scale sensitivities of less than $50\ \mu\text{A}$, whereas taut-band meters are available with full-scale sensitivity of only $2\ \mu\text{A}$. D' Arsonval movements are more easily damaged than taut-band movements because of their jewel-bearing construction. Even a short-fall to the floor or bench top is sufficient in most cases to break the bearings or dislodge the armature from the bearing cups. The rubber band in a taut-band movement can snap, but this occurs far less frequently than does bearing damage in d' Arsonval movements. While abusing either type of meter is bad business, the taut-band movement is more forgiving and can absorb more punishment than most d' Arsonval movements.

Types of Analog Meters

The two basic meter movements are available in an almost endless variety of sizes and shapes, but we can classify meter types by the form of scale that is used. Figure 1-4 shows three different configurations.

The meter in Fig. 1-4(a) is the type normally encountered. The zero-current mark is on the left side of the scale, and the pointer deflects to the right when a current flows (assuming the meter is hooked into the circuit correctly!). If the meter is a d~ type, then the input terminals of the meter will be marked positive (+) and negative (-) (the negative is sometimes not marked). The terminals must not be reversed, or the pointer may be damaged when it attempts to deflect further to the left. Figure 1-4(b) shows an example of a zero-center de microammeter; the zero-current mark on this type of meter is in the exact center of the scale. Current flowing in one direction will deflect the pointer to the left of center, while current flowing in the opposite direction will deflect the pointer to the right.

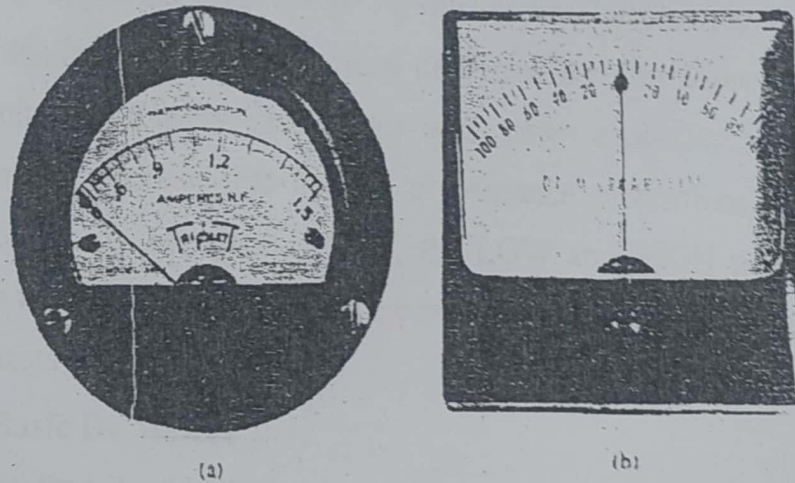


Fig. (4) (a) Thermocouple RF ammeter. (b) zero-center de microammeter.

When the polarity markings on the input terminals are observed, a positive current deflects the pointer to the right, while a negative current deflects the pointer to the left.

The type of meter movement shown in Fig .1-4(c) is an edgewise panel meter. Its design allows conservation of area on the front panels of equipment in which the meter is used. This saving is at the expense of increased depth, but that is often considered an acceptable engineering trade-off. Edgewise meters are available in both left-zero and center-zero models.

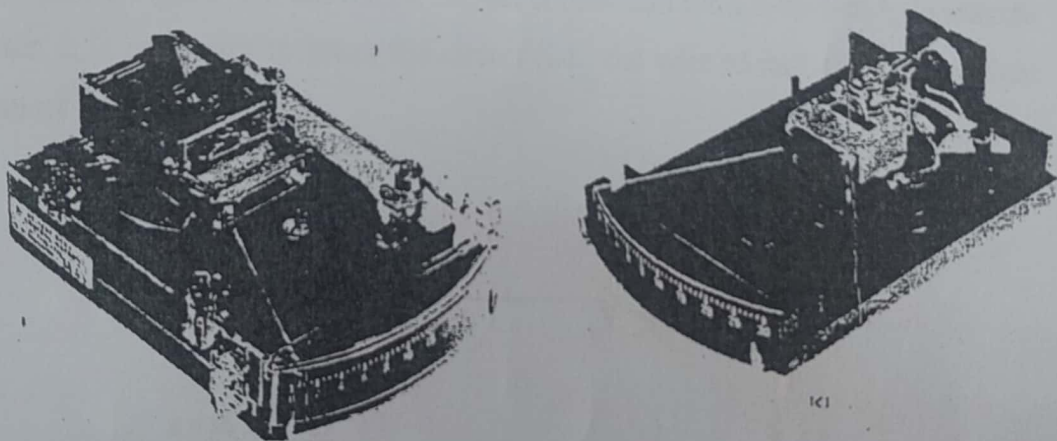


Fig. (4) Cont. (c) Two views of an edgewise panel meter movement.

The edgewise panel meter movement has internal photoelectric alarm sensors that indicate when the pointer deflection is below a low limit or above a high limit. Other meter movements with alarm capability have an internal potentiometer element that outputs a resistance proportional to the pointer deflection. In optical alarm sensors, the LOW and HIGH points are set by movable tabs on the meter face. In the resistance type of alarm, external voltage comparators are usually used.

Using Basic De Meters

There are three basic rules for using de current meters:

1. Connect the meter in series with the load or circuit in which the current is being measured.
2. Use a meter with a full-scale current rating that is greater than the maximum current expected.
3. Use a meter that has an internal resistance that is low compared with the resistance in which it is being used (i.e., $<1:10$).

A current meter is always connected in series with the load! There are never any exceptions. *Failure to observe this rule may result in permanent, irreversible damage to the instrument.* When a meter is used to measure current in a branch of a larger circuit, then the meter is to be connected in series with that branch. Figure 1-5 shows the correct connection for a 0-to-1-mA de current meter. It is necessary to break the circuit (i.e., the wire to resistor R_1) and then connect the meter in series.

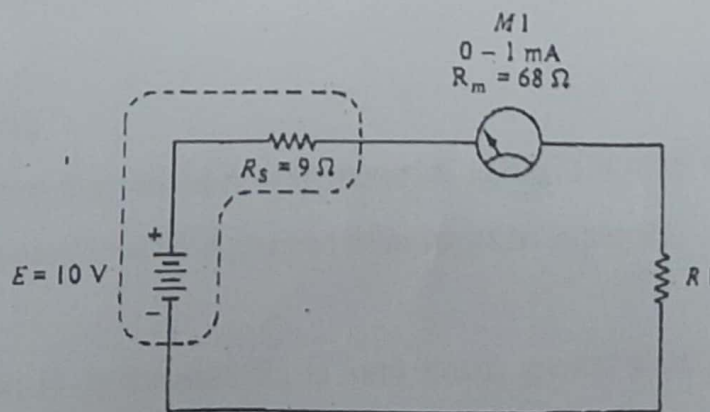


Fig. (1-5) Circuit with 0-to-1 -mA milliammeter.

The current flowing in meter M_1 is given by

$$I_{M1} = \frac{E}{R_s + R_m + R_l} \quad (1-1)$$

where I_{M1} = the meter current in amperes (A)

E = the open-circuit voltage in volts (V)

R_s = the power supply source resistance in ohms (Ω)

R_l = the load resistance in ohms (Ω).

In most cases, R_s and R_m are very small with respect to R_l and so are considered negligible. If this is true, then Equation (1-1) is reduced to

$$I_{M1} = \frac{E}{R_l} \quad (1-2)$$

Example : 1-2

Find the current flowing in meter M_1 in Fig. 1-5 if M_1 when $R_l = 15 \text{ k}\Omega$.

Solution :

Since R_l is very much greater than R_s or R_m , use Equation (1-2):

$$\begin{aligned} I_{M1} &= \frac{E}{R_l} \\ &= \frac{10 \text{ V}}{1.5 \times 10^{-4}} \\ &= 6.7 \times 10^{-4} \text{ A} = 0.67 \text{ mA} \end{aligned}$$

Example : 1-2

Find the current that would flow in meter M_1 in Fig. 1-5 if M_1 were *incorrectly* connected in parallel with R_l instead of in series as is proper.

Solution

Use Equation (1-1) because R_l is very much greater than R_s and R_m , so its parallel effect is negligible. Delete R_l from the equation:

$$\begin{aligned}
 I_{M1} &= \frac{E}{R_s + R_m} \\
 &= \frac{10 \text{ V}}{(0 + 68) \Omega} = 0.13 \text{ A} = 130 \text{ mA}
 \end{aligned}$$

Connecting M_1 across the load instead of in series results in a current that is 130 times too high for the full-scale rating. This current will probably destroy the instrument.

Rule number 3 can become troublesome in low-voltage, low-resistance circuits, where the meter's coil resistance (R_m) is a significant portion of the load resistance. Neglecting R_s , let's see what would happen if R_l were reduced to 150Ω and E were only 100 mV (i.e., 0.1 V). By Ohm's law, if R_m were negligible, then we would expect a current of

$$\begin{aligned}
 I &= \frac{E}{R_l} \\
 &= \frac{0.1 \text{ V}}{150 \Omega} = 0.00067 \text{ A} = 0.67 \text{ A}
 \end{aligned}$$

But with R_m being a significant portion of R_l , the actual current is $(0.1 \text{ V})/(68 + 150 \Omega)$, or 0.46 mA.

No measuring or testing instrument should be used if it significantly alters the circuit performance, and this is the case in the example above. A 31% meter reading error was created by an unwise selection of the meter used to make the measurement. In both cases, above, we expected to find $0.67 \mu\text{A}$, but in the latter case the reading was only $0.46 \mu\text{A}$.

OBTAINING HIGHER CURRENT SCALES

The basic de meter movement has a single current scale (e.g., 0 to 1 μA , 0 to 200 μA , etc.), but it can be used to measure larger currents if a **shunt resistor** in parallel with the meter is used (see Fig. 1,-6).

The wire diameter required for the coil in high-current meter movements is excessive and proves impractical, so it has become standard practice to use a low-range meter movement and the shunt resistor. In some cases the shunt resistor is mounted inside the meter case, whereas in others it is located outside the case (usually mounted directly to the meter's input terminals). An

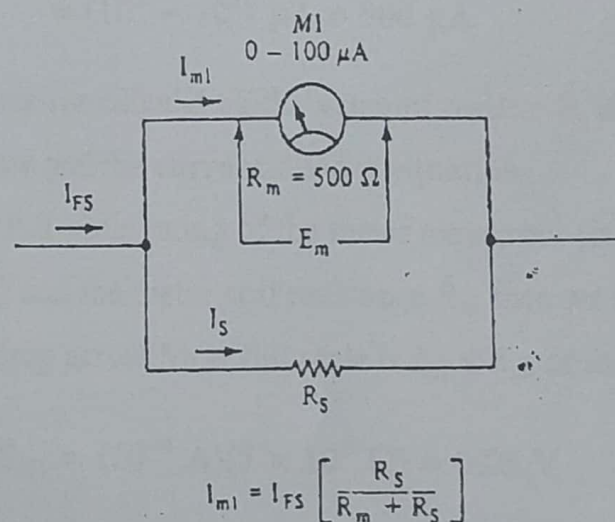


Fig. (1-6) Principle of the shunt resistor to increase current range.

Example using the shunt resistor is given in Fig. 3-6. The total full-scale current of the meter with shunt in place is given by

$$I_{FS} = I_{M1} + I_S \quad (1-3)$$

where I_{FS} = the total current

I_{M1} = the current flowing in the meter

I_S = the current flowing in the shunt

Example (1-3) : The circuit of Fig. 1 -6 is used to measure 1 mA full scale. How much current at full scale flows in (a) $M2$ and (b) R_s ?

Solution

Solve equation (1-3) for I_s :

$$I_{FS} = I_{M1} + I_s$$

Therefore :

$$\begin{aligned} I_s &= I_{FS} - I_{M1} \\ &= \left[1 \text{ mA} \times \frac{10^3 \mu\text{A}}{1 \text{ mA}} \right] - 10^2 \mu\text{A} \\ &= (10^3 - 10^2) \mu\text{A} = 900 \mu\text{A} \end{aligned}$$

There are two methods for calculating the value of resistor R_1 if R_m , I_{M1} and I_{FS} are known: **Ohm's law** and the **current divider equation**.

If we know both the full-scale rating of the meter movement [in Example 1-3 it is $100 \mu\text{A}$ (Fig. 1 -6)] and the meter coil resistance R_m , then we know by Ohm's law that the voltage drop across M_1 at full scale is $I_{M1} \times R_m$, or in this case

$$E_{M1} = (10^{-4} \text{ A})(5 \times 10^2 \Omega) = 0.05 \text{ V}$$

Since R_s is connected in parallel with M_1 , the same voltage is also across R_s . If we calculate I_s [Eq. (1-3)] and use Ohm's law, we can find the value required of R_s . In the case of Example 1-3, in which $900 \mu\text{A}$ flows in R_s ,

$$\begin{aligned} R_s &= \frac{E_{M1}}{I_s} \\ &= \frac{0.05 \text{ V}}{9 \times 10^{-4} \text{ A}} = 55.56 \Omega \end{aligned}$$

The step-by-step procedure for using Ohm's law to determine the value of the shunt resistor is as follows:

1. Determine I_{FS} , I_{M1} and R_m .
2. Calculate I_s , using Equation (1-3) .
3. Calculate E_{M1} , using Ohm's law.
4. Use Ohm's law again to calculate R_s from E_{M1} and I_{FS} or, in equation form,

$$R_s = \frac{I_{M1} R_{M1}}{I_{FS} - I_{M1}} \quad (1-4)$$

The current divider method takes advantage of Kirchhoff's current law (KCL), which tells us that the current in R_s is a fraction of I_{FS} :

$$I_s = I_{FS} \times \left(\frac{R_m}{R_m + R_s} \right) \quad (1-5)$$

Example 1.-4

A 0-to 50 μ A de meter movement, with a coil resistance R_m of 1250 Ω , is used with a shunt resistor to measure a full-scale current of 500 μ A. Calculate the value of R_s .

Solution

Solve Equation (1- 5) for R_s :

$$I_S = I_{FS} \times \left[\frac{R_m}{R_m + R_S} \right]$$

$$(500 \mu\text{A} - 50 \mu\text{A}) = (500 \mu\text{A}) \times \left[\frac{1250 \Omega}{(1250 + R_S \Omega)} \right]$$

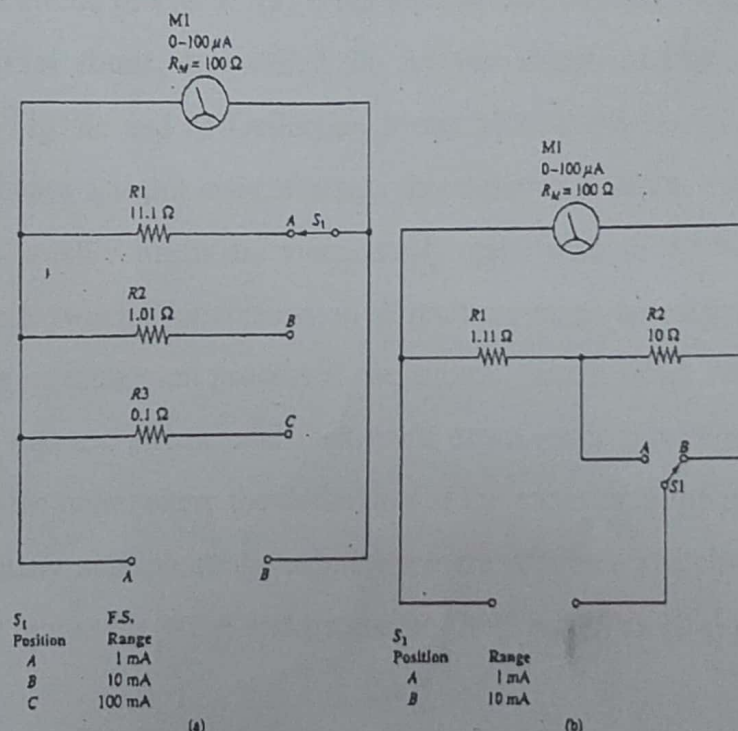
$$\frac{450 \mu\text{A}}{(500 \mu\text{A})(1250 \Omega)} = \frac{1}{(1250 + R_S) \Omega}$$

By algebraically rearranging the expression, we obtain

$$R_S = \left[\frac{(500)(1250)}{450} \Omega \right] - 1250 \Omega$$

$$= (1389 - 1250) \Omega = 139 \Omega$$

Although each method for determining the value of the shunt resistor has its own advocates, either method will yield the same answer. You are encouraged to work Exampld-4 using the Ohm's law method.



(a) Three-scale, multiple-range ammeter; (b) universal shunt, multiple-range ammeter

If a switch is provided that permits the selection of any of several different-value shunt resistors, then it is possible to make a **multiple-range ammeter**.

Each different shunt resistor is used to set the full-scale meter current to a different value, resulting in different switch-selectable current ranges being covered. Thus, a 0-to-100- μA dc microammeter can be used for full-scale readings of, for example, 100 μA , 500 μA , 1000 μA (1 mA), 10 mA, 50 mA, 100 mA, 500 mA, and 1 A. An example of a simple three scale, multiple-range milliammeter is shown in Fig. 1-7(a).

A flaw in this design is that during the instant when the switch is being transferred from one position to another, the meter movement is in the circuit *without a shunt* and therefore is subject to damage. A solution to the problem is to use a "make contact before breaking contact" type of switch to select the various shunt resistors.

The circuit of Fig. 1-7(a) is not considered optimum. A superior circuit is the **universal shunt**, also called the **Ayrton shunt**, multiple-range ammeter shown in Fig. de and ac Deflection Meter Movements .1-7(b). In this circuit there is always a shunt resistor across the meter movement, even during times when the switch contacts are momentarily open, without the need for a make-before-break switch. Nonetheless, in all multiple-range ammeters it is preferable to start the measurement process at the highest current range before connecting the meter into the circuit, and then work down (once power is applied to the circuit) to the point where the deflection of the meter is in the upper half of the scale. In many multiple-range ammeters there will be a switch-selectable input for current ranges tip to 1 A and a separate "hot" input for a 10-; range.

Some multiple-range ammeters are autoranging. In these instruments an electronic switch automatically sequences from the highest current range to the lowest current range until a setting is found that results in a reading that is less than full scale. This feature is more common in digital ammeters but is also found in at least a few analog designs.

Voltage Measurements From DC Current Meters

Voltage can be measured on a dc current meter if a multiplier resistor is connected in *series* with the meter movement. This type of circuit is shown in Fig. 1-8(a), with the equivalent circuit shown in Fig. 1-8(b).

Input voltage E will set up current I_1 in the circuit of Fig. 1-8(a) that has a value of

$$I_1 = \frac{E}{R_m + R_{mx}} \quad (1-6)$$

where I_1 = the current in amperes (A)

E = the applied potential in volts (V)

R_m = the meter resistance in ohms (Ω)

R_{mx} = the multiplier resistance in ohms (Ω).

We may calculate the multiplier resistance needed for any given full-scale rating of MI and then solve Equation (1-6) for R_{mx} .

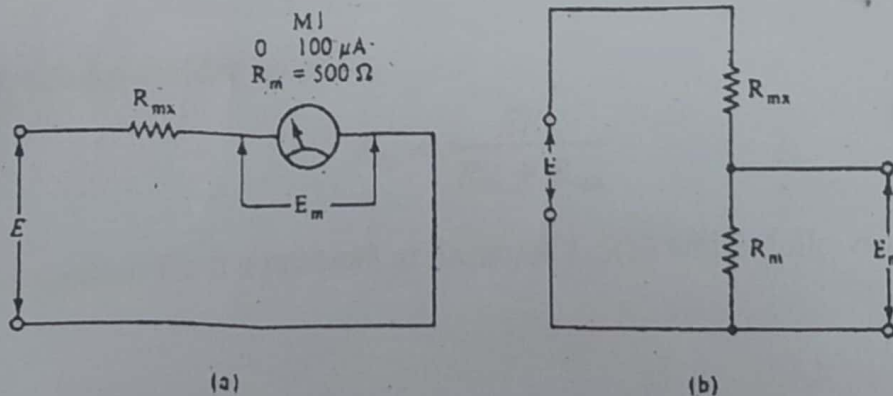


Fig. (1-8) (a) Use of multiplier resistor to make an ammeter read in volts. (b) equivalent circuit.

• Example :1-5(a)

Calculate the value of the multiplier resistor R_{mx} in Fig . 1-8(a) for a full-scale potential E of 10 V.

Solution

Solve Equation (1 -6) for R_{mx} :

$$I_1 = \frac{E}{R_m + R_{mx}}$$

$$10^{-4} \text{ A} = \frac{10 \text{ V}}{(500 + R_{mx}) \Omega}$$

$$10^{-5} \text{ A} = \frac{1 \text{ V}}{(500 + R_{mx}) \Omega}$$

By rearranging the expression, we obtain

$$R_{mx} + 500 \Omega = 10^5 \Omega$$

$$R_{mx} = 10^5 \Omega - 500 \Omega = 99,500 \Omega$$

We may also use the **voltage divider equation** to calculate the value of R_{mx} .

We know that the full-scale voltage drop across meter M1 will be equal to $I_1 \times R_1$. Thus,

$$\begin{aligned} E_m &= I_1 \times R_m \\ &= (10^{-4} \text{ A})(5 \times 10^2 \Omega) = 0.050 \text{ V} \end{aligned}$$

So by the voltage divider equation,

$$E_m = \frac{ER_m}{R_m + R_{mx}} \quad (1-7)$$

We find a solution for the problem of Example 1-5(b) which follows.

Example :1-5(b)

Work the problem of Example .1-5(a) using the voltage divider equation method.

Solution :

$$E_m = \frac{ER_m}{R_m + R_{mx}}$$

$$0.050 \text{ V} = \frac{(10 \text{ V})(500 \Omega)}{(500 + R_{mx}) \Omega}$$

DC and AC Deflection Meter Movements With a little algebraic manipulation, we find

$$R_{mx} + 500 = \frac{(10)(500)}{0.050}$$

$$R_{mx} = (100,000 - 500) \Omega = 99,500 \Omega$$

The resistance is the same as before, as expected.

Voltmeter Sensitivity (ϕ)

The sensitivity (ϕ) of a voltmeter is specified in terms of ohms per volt (Ω/V) and is dependent upon the full-scale current range of the basic de meter movement used to make the voltmeter. The sensitivity can be found by taking the reciprocal of the meter's full scale current:

$$\phi = \frac{1}{I_{FS}} \quad (1-8)$$

Table (1-1) gives the sensitivity ratings for de voltmeters made by using various values of full-scale meter ratings .

DC VOLTMETER RESISTANCE

The **resistance** of a dc voltmeter is the product of its full-scale rating and the **sensitivity** rating:

$$R_m = E_{FS} \times \phi \, \Omega \quad (1-9)$$

where R_m = the resistance of the voltmeter in ohms (Ω)

E_{FS} = the full-scale rating of the meter in volts (V)

ϕ = the sensitivity of the meter in ohms per volt (Ω / V)

Table (1-1) Sensitivity ratings for dc voltmeters

Full-Scale Meter Current	Sensitivity (ϕ)
1 mA	1000 Ω / V
100 μA	10 k Ω / V
50 μA	20 k Ω / V
20 μA	50 k Ω / V
10 μA	100 k Ω / V

Example (1-6) :

Calculate the input resistance of a dc voltmeter such as in Fig. 1-8(a) on the 0-500V dc scale if the sensitivity is 10,000 Ω/V .

Solution

$$\begin{aligned} R_m &= E_{FS} \times \phi \\ &= (500 \, V)(10^4 \, \Omega/V) = 5 \times 10^6 \, \Omega = 5 \, M\Omega \end{aligned}$$

Using Voltmeters

Voltage is the *difference in electrical potential between two points* in a circuit. Therefore, to measure voltage, connect the voltmeter *across the load*, that is, in *parallel* with the load or circuit branch that you want to measure. Three basic rules apply:

1. Connect the voltmeter in parallel with the load (see Fig. 1-9).
2. Select a voltmeter that has a full-scale range that is greater than the highest potential expected.
3. Make sure that the voltmeter has an input resistance (Section 1-11) that is *very high* (e.g., $> 100:1$) compared with the circuit resistance.

The voltmeter is a high-impedance device, so any attempt to connect it in series. With the circuit will greatly increase circuit resistance, causing improper operation.

Problems of circuit loading will occur if rule number 3 is not obeyed. Indeed, there may be circuits where the $> 100:1$ rule is not enough (electronic millimeters will be needed in those cases. Consider what will happen if a circuit is loaded down by the voltmeter resistance [(Fig. 1- 10(a))]. From the voltage divider equation we know that E_2 should be

$$\begin{aligned} E_2 &= \frac{ER_2}{R_1 + R_2} \\ &= \frac{(10 \text{ V})(1 \text{ M}\Omega)}{1 \text{ M}\Omega + 0.5 \text{ M}\Omega} \\ &= \frac{10 \text{ V}}{1.5} = 6.67 \text{ V} \end{aligned}$$

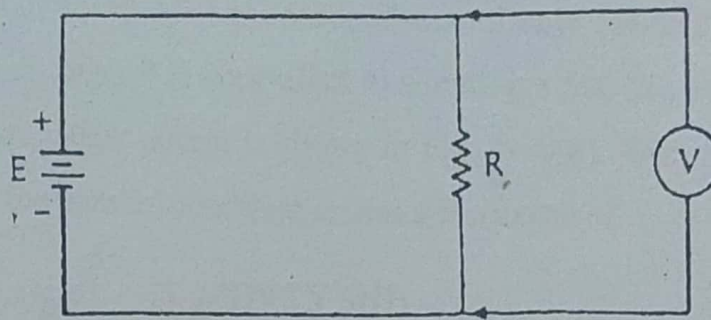
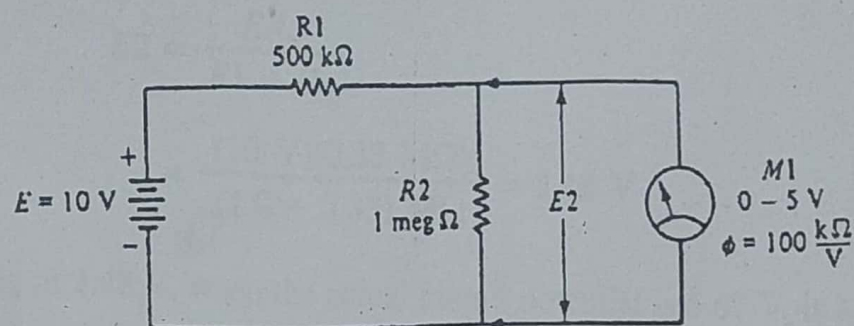
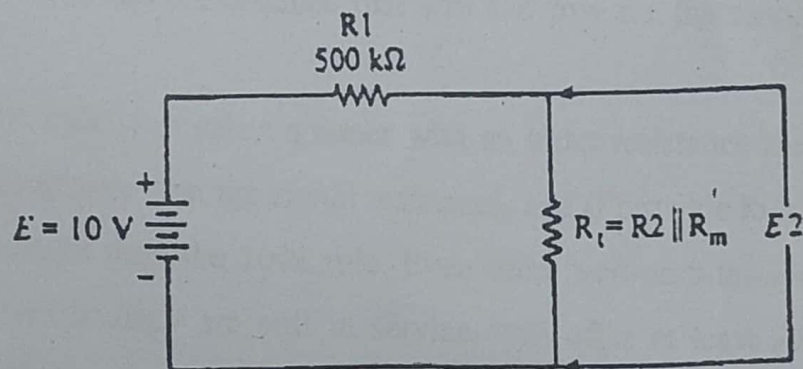


Fig. (1-9) A voltmeter is always connected in parallel with the load.

DC and AC Deflection Meter Movements



(a)



(b)

Fig. (1-10) (a) Voltmeter impedance can cause errors due to circuit loading;
(b) equivalent circuit.

But at 0 to 5 V full scale; M_1 has an impedance of only 500 k Ω . When M_1 is connected across R_2 , then it is equivalent to shunting a 500-k Ω . resistor across resistor R_2 ; the equivalent circuit is shown in Fig. 1-10(b). With R_m in parallel with R_2 ($R_m \parallel R_2$), the parallel combination has a resistance of

$$R_{eq} = \frac{(1 \text{ M}\Omega)(0.5 \text{ M}\Omega)}{1 \text{ M}\Omega + 0.5 \text{ M}\Omega} = 0.33 \text{ M}\Omega$$

Again using the voltage divider equation, but this time with the equivalent resistance R_{eq} substituted for R_2 , we find that voltage E_2 , as indicated on the meter, is

$$\begin{aligned} E_2 &= \frac{ER_{eq}}{R_1 + R_{eq}} \\ &= \frac{(10 \text{ V})(0.33 \text{ M}\Omega)}{(1.0 + 0.33) \text{ M}\Omega} = 2.48 \text{ V} \end{aligned}$$

A reading of 2.48 V, when the actual circuit potential is 6.67 V, is a 63% error, and that error is caused solely by the fact that the measurement was made with a voltmeter that had a resistance that was too low for the circuit being tested.

The general rule is to select a meter with an input resistance that is not less than 10 times greater than the circuit resistance, and if possible to make the meter resistance more than the 100x rule. Even older volt-ohm-milliammeter instruments, of which many are still in service, will offer at least 20 k Ω /V resistance, and many are found with 100 k Ω /V.

Beware, however, that many 1000 Ω /V instruments are still in service, and they can be deadly to good measurements. (Indeed, they may upset the circuit enough to cause secondary damage!)

A valid question to ask is, "Why use a passive instrument at all?" In most cases an active instrument such as a digital multimeter is the instrument of choice; but where high electrical or varying magnetic fields are encountered, the

electronic voltmeter suffers from input biasing due to electromagnetic interference. People who work on radio transmitters find this problem quite often. The passive meter circuitry of the volt-ohm-millimeter is far more immune to EMI than even most of the best active meters. Unfortunately, the kind of filtering methods that are used to eliminate EMI also sometimes distort waveforms being measured.

DC ANALOG OHMMETERS

Electrical resistance is measured using an instrument called an **ohmmeter**. According to Ohm's law, the current in a circuit is directly proportional to the applied voltage (E) and inversely proportional to the resistance (R) of the circuit; that is, $I = E/R$. That relationship can be used to find an unknown resistance (R_x), if a stable voltage is available. The circuit for a simple series **ohmmeter** is shown in Fig. 1-11(a). There are three resistances in this circuit: R_m is the internal resistance of the voltmeter (M), R_1 is the resistance of a zero calibration control potentiometer, and R_x is the unknown resistance.

The scale for ohmmeters is reversed in direction from the voltage and current scales. That is, the highest values are on the left side of the scale, while the lowest values are on the right. In the ohmmeter of Fig. 1-11(a), the zero point occurs when maximum current flows (which puts the meter on the right side), that is, when terminals A and B are shorted together. Typical operation of the analog ohmmeter requires that the leads be shorted together with the pointers zeroed with the calibration control. On modern digital voltmeters, by the way, the zeroing is automatic. A limitation of the series ohmmeter circuit is that it does not work well at low resistances ($R_x < 100\Omega$). For lower resistance ranges the parallel ohmmeter, also called the

electronic voltmeter suffers from input biasing due to electromagnetic interference. People who work on radio transmitters find this problem quite often. The passive meter circuitry of the volt-ohm-millimeter is far more immune to EMI than even most of the best active meters. Unfortunately, the kind of filtering methods that are used to eliminate EMI also sometimes distort waveforms being measured.

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$$I_m R_m = I_s R_s$$

$$= (I_F - I_m) R_s$$

$$I_m R_m + I_m R_s = I_F R_s$$

$$I_m (R_s + R_m) = I_F R_s$$

$$I_m = I_F \frac{R_s}{R_s + R_m}$$

$$\begin{array}{r} 1000 \\ 1000 \overline{) 1000} \\ \underline{1000} \\ 0 \end{array}$$